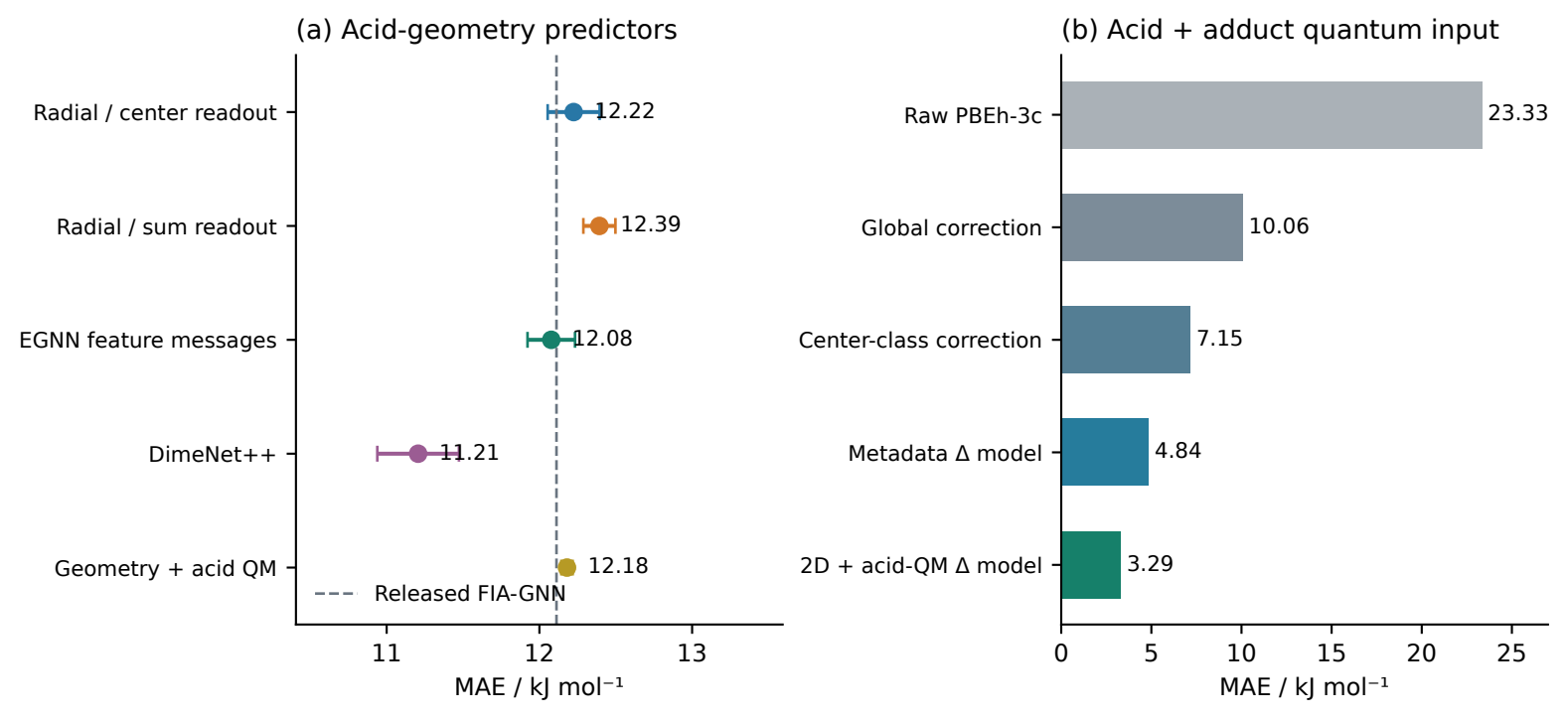
(a) Acid-geometry predictors
Radial / center readout
12.22
Radial / sum readout
12.39
EGNN feature messages
12.08
DimeNet++
11.21
Geometry + acid QM
12.18
Released FIA-GNN
11
12
13
MAE / kJ mol⁻¹
(b) Acid + adduct quantum input
Raw PBEh-3c
23.33
Global correction
10.06
Center-class correction
7.15
Metadata Δ model
4.84
2D + acid-QM Δ model
3.29
0
5
10
15
20
25
MAE / kJ mol⁻¹